Question 1:
Which of the following functions can be used to check if a file exists?

A. os.path.isfile( )
B. os. path.exists( )
C. os.isFile( )
D. os.path.isFile( )

Question 2:
What is the expected output of the following code?

```
data1 = '1', '2'
data2 = ('3', '4')
print(data1 + data2)
```

A. The code is erroneous.
B. (1, 2, 3, 4)
C. ['1' , '2' , '3' , '4']
D. ('1' , '2' , '3' , '4')

Question 3:
ASCII is:

A. a predefined Python variable name
B. short for American Standard Code for Information Interchange
C. a character name
D. a standard Python module name

Question 4:
How many arguments can the print() function take?

A. Not more than seven arguments.
B. Just one argument.
C. Any number of arguments (including zero).
D. Any number of arguments (excluding zero).

Question 5:
What is the expected output of the following code?

```
def func(x):
    if x % 2 == 0:
        return 1
    else:
        return

print(func(func(2)) + 1)
```

A. The code is erroneous.
B. 2
C. 1
D. None

Question 6:
How many stars will the following snippet print to the monitor?

```
for i in range(1):
    print('*')
else:
    print('*')
```

A. one
B. zero
C. two
D. three

Question 7:
The two basic, mutually exclusive, file open modes are named:

A. text and image
B. binary and text
C. binary and ternary
D. None of the above.

Question 8:
You want to write a program that asks the user for a value. For the rest of the program you need a whole number, even if the user enters a decimal value. What would you have to write?

A. num = str(input('How many do you need? '))
B. num = int(float(input('How many do you need ?')))
C. num = float(input('How many do you need? '))
D. num = int('How many do you need? ')

Question 9:
Which of the following commands can be used to read the next line from a file?

A. readlines( )
B. readline( )
C. read(n)
D. read( )

Question 10:
What is the expected output of the following code?

```
list1 = [3, 7, 23, 42]
list2 = [3, 7, 23, 42]
print(list1 is list2)
print(list1 == list2)
```

A.	1 | True 2 | False
B.	1 | False 2 | True
C.	1 | True 2 | True
D.	1 | False 2 | False

Question 11:
What is the expected output of the following code?

```
x = 42

def func():
    global x
    print('1. x:', x)
    x = 23
    print('2. x:', x)

func()
print('3. x:', x)
```

A.	1 | 1. x: 42 2 | 2. x: 23 3 | 3. x: 42
B.	1 | 1. x: 42 2 | 2. x: 23 3 | 3. x: 23
C.	None of the above.
D.	1 | 1. x: 42 2 | 2. x: 42 3 | 3. x: 42

Question 12:
The following is a program to validate customer numbers.

```
customer_number = input('Enter the employee number (dd-ddd-dddd): ')
parts = customer_number.split('-')
valid = False
if len(parts) == 3:
  if len(parts[0]) == 2 and len(parts[1]) == 3 and len(parts[2]) == 4:
    if parts[0].isdigit() and parts[1].isdigit() and parts[2].isdigit():
      valid = True
print(valid)
```

The number may only contain numbers and dashes. The number must have the right format (dd-ddd-dddd). What is true about this program?

A. The program works properly.
B. There will be a SyntaxError.

C. There will be no error but there will be an unwanted result.
D. There will be an AttributeError.

Question 13:
A built-in function is a function which ...

A. comes with Python, and is an integral part of Python.
B. has to be imported before use.
C. has been placed within your code by another programmer.
D. is hidden from programmers.

Question 14:
What is the expected output of the following code?

```
nums = [3, 7, 23, 42]
alphas = ['p', 'p', 'm', 'j']

print(nums is alphas)
print(nums == alphas)

nums = alphas

print(nums is alphas)
print(nums == alphas)
```

A.	1 | False 2 | True 3 | False 4 | True
B.	1 | False 2 | True 3 | True 4 | True
C.	1 | True 2 | False 3 | True 4 | False
D.	1 | False 2 | False 3 | True 4 | True

Question 15:
What is the expected output of the following code?

```
x = 1
print(++++x)
```

A. 3
B. 2
C. 1
D. 4

Question 16:
What is the expected result of the following code?

```
from datetime import timedelta

delta = timedelta(weeks=1, days=7, hours=11)
print(delta * 2)
```

A. The code will raise an exception.
B. 7 days, 22:00:00
C. 2 weeks, 14 days, 22 hours
D. 28 days, 22:00:00

Question 17:
Consider the following Python code:

```
name = 'Peter'
age = 23
flag = True
```

What are the types of the variables name, age and flag?

A. str, int, int
B. int, bool, char
C. float, bool, str
D. str, int, bool

Question 18:

What is the expected output of the following code?

```
x = 2
y = 6
x += 2 ** 3
x //= y // 2 // 3
print(x)
```

A. 10
B. 0
C. 9
D. 11

Question 19:
What is the expected output of the following code?

```
data = [1, 2, 3, None, (), [], ]
print(len(data))
```

A. 4
B. 5
C. 3
D. 6
E. The code is erroneous.

Question 20:
What is the expected output of the following code?

```
def func():
    try:
        return 1
    finally:
        return 2

res = func()
print(res)
```

A. 2
B. 1
C. The code is erroneous.
D. None of the above.

Question 21:
What is the data type of x, y, z after executing the following snippet?

```
x = 23 + 42
y = '23' + '42'
z = '23' * 7
```

A. int, str, int

B. x is int, y and z are invalid declarations
C. int, int, int
D. int, str, str

Question 22:
What is the expected output of the following code?

```
x, y, z = 3, 2, 1
z, y, x = x, y, z
print(x, y, z)
```

A. 3 2 1
B. 1 2 3
C. 1 2 2
D. 2 1 3

Question 23:
What is the expected output of the following code?

```
data = [1, 2, [3, 4], [5, 6], 7, [8, 9]]
count = 0

for i in range(len(data)):
    if type(data[i]) == list:
        count += 1

print(count)
```

A. 6
B. 3
C. 9
D. The code is erroneous.

Question 24:
You want to print the sum of two number. What snippet would you insert in the line indicated below:

```
x = input('Enter the first number: ')
y = input('Enter the second number: ')
#  insert your code here
```

A. print (‘ The Result is ‘ + str(int(x + y)))
B. print (‘ The Result is “ + str(int(x) + int (y)))
C. print (‘ The Result is ‘ + (int(x) + int(y)))
D. print (‘ The Result is ‘ + (int(x+y)))

Question 25:
What is the expected output of the following code?

```
data = {'name': 'Peter', 'age': 30}
person = data.copy()
print(id(data) == id(person))
```

A. 0
B. False
C. True
D. 1

Question 26:
What is the expected output of the following code?

```
def func(x, y):
    if x == y:
        return x
    else:
        return

func(x, y=1)
print(func(0, 3))
```

A. 0
B. The code is erroneous.
C. 1
D. 3

Question 27:
You have the following file.

```
index.py:
from sys import argv
sum = 0
for i in range(2, len(argv)):
    sum += float(argv[i])
print(
    "The average score for {0} is {1:.2f}"
    .format(argv[1], sum/(len(argv)-2))
)
```

You want the following output.

```
The average score for Peter is 200.00
```

Which command do you have to execute in the command line?

A. The code is erroneous.
B. python index.py Peter 100
C. python index.py Peter 100 200 300
D. python index.py Peter 100 200

Question 28:
What would you insert instead of ??? so that the program prints True to the monitor?

```
x = 'Peter'
y = 'Peter'
res = ???
print(res)
```

A. x is not y
B. x != y
C. x is y
D. x < y

Question 29:
If x is a file opened in read mode, what will the following line do?

```
data = x.read(1)
```

A. Read 1 line from the file.
B. Read 1 kilobyte from the file.
C. Read 1 buffer from the file.
D. Read 1 character from the file.

Question 30:
You need a list of seven numbers of random integer values from 1 (inclusive) to 7 (inclusive). Which of the following code snippets should you use?

A.	1 | import random 2 | nums = [random.randint(1, 8) for i in range(1, 8)]
B.	1 | import random 2 | nums = random.randint(1, 7)
C.	1 | import random 2 | nums = [random.randint(1, 7) for i in range(1, 8)]
D.	1 | import random 2 | nums = random.randrange(1, 7)

Question 31:
What is the expected output of the following code?

```
def func(x=2, y=3):
    return x * y

print(func(y=2))
```

A. 6
B. The code is erroneous.
C. 2
D. 4

Question 32:
What is the expected output of the following code?

```
data = {'Peter': 30, 'Paul': 31}
print(list(data.keys()))
```

A. [‘Peter’ , ‘Paul’]
B. [‘ Peter’ : 30, ‘Paul’ : 31]
C. (‘Peter’ , ‘Paul’)
D. (‘Peter’ : 30, ‘Paul” : 31)

Question 33:
What is the expected output of the following code?

```
try:
    print("5" / 0)
except ArithmeticError:
    print("arith")
except ZeroDivisionError:
    print("zero")
except:
    print("some")
```

A. 0
B. zero
C. arith
D. some

Question 34:
What is the expected output of the following code?

```
class Ex(Exception):
    def __init__(self, msg):
        Exception.__init__(self, msg + msg)
        self.args = (msg,)

try:
    raise Ex('ex')
except Ex as e:
    print(e)
except Exception as e:
    print(e)
```

A. ex
B. exex

C. The code is erroneous.
D. An empty line.

Question 35:
What is the expected output of the following code?

```
print(2 ** 3 ** 2 ** 1)
```

A. The code is erroneous.
B. 16
C. 16.0
D. 512

E. 64
F. 128.0

Question 36:
What is the expected output of the following code?

```
data = (1, 2, 4, 8)
data = data[-2:-1]
data = data[-1]
print(data)
```

A. 44
B. (4)
C. 4
D. (4,)

Question 37:
What is the expected output of the following code?

```
try:
    raise Exception
except BaseException:
    print('1', end='')
else:
    print('2', end='')
finally:
    print('3', end='')
```

A. 1
B. 12
C. 13
D. 23

Question 38:
What is the expected output of the following code?

```
x = '\\\'
print(len(x))
```

A. 1
B. 2
C. The code is erroneous.
D. 3

Question 39:
Select the true statements. Choose two.

A. The finally branch of the try statement is always executed.
B. The finally branch of the try statement may be executed if special conditions are met.
C. The args property is a tuple designed to gather all arguments passed to the class constructor.
D. You cannot define new exceptions as subclasses derived from predefined exceptions.

Question 40:
What is the expected output of the following code?

```
def func(data):
    data = [7, 23, 42]
    print('Function scope: ', data)

data = ['Peter', 'Paul', 'Mary']
func(data)
print('Outer scope: ', data)
```

A.	None of the above.
B.	1 | Function scope: [7, 23, 42] 2 | Outer scope: [7, 23, 42]
C.	1 | Function scope: [‘Peter’ , ‘Paul’ , ‘Mary’] 2 | Outer scope: [‘Peter’ , ‘Paul’ , ‘Mary’]
D.	1 | Function scope: [7, 23, 42] 2 | Outer scope: [‘Peter’ , ‘Paul’ , ‘Mary’]

Question 41:
You are developing a Python application for an online product distribution company. You need the program to iterate through a list of products and escape when a target product ID is found.

```
productIdList = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
index = 0

??? index < 10:
    print(productIdList[index])
    if productIdList[index] == 6:
        ???
    else:
        index += 1
```

What would you insert instead of ??? and ???

A.	1 | break 2 | while
B.	1 | for 2 | break
C.	1 | if 2 | break
D.	1 | for 2 | while
E.	1 | while 2 | break
F.	1 | while 2 | for

Question 42:
What is the expected output of the following code if the user enters 2 and 4?

```
x = int(input())
y = int(input())
print(x + y)
```

A. 24
B. 6
C. 4
D. 2

Question 43:
What is the expected output of the following code?

```
print(3 * 'abc' + 'xyz')
```

A. abcabcxyzxyz
B. abcxyzabcxyzabcxyz
C. abcabcabcxyz
D. 3abcxyz

Question 44:
Consider the following code.

```
data = ['Peter', 'Paul', 'Mary', 'Jane']
res = 0
```

Which of the following code snippets will expand the code, so that 100 will be printed to the monitor? Choose two.

A.	1 | for i in (‘Peter’ , ‘Steve’ , ‘Jane’): 2 | if i not in data: 3 | res += 50 4 | print(res)
B.	1 | for i in (‘Peter’ , ‘Steve’ , ‘Jane’): 2 | if i not in data: 3 | res += 100 4 | print(res)
C.	1 | for i in (‘Peter’ , ‘Steve’ , ‘Jane’): 2 | if i not in data: 3 | res += 100 4 | print(res)
D.	1 | for i in (‘Peter’ , ‘Steve’ , ‘Jane’): 2 | if i not in data: 3 | res += 50 4 | print(res)

Question 45:
Consider the following code.

```
1  nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
2  x = 0
3  while x < 10:          # Line 3
4      print(nums(x))     # Line 4
5      if nums[x] = 7:    # Line 5
6          break          # Line 6
7      else:              # Line 7
8          x += 1         # Line 8
```

You want to print the numbers 1 to 7 to the monitor. But the code does not work. What do you have to change? Choose two.

A.	x = x + 1	# Line 8
B.	while (x < 10):	# Line 3
C.	if nums[x] == 7:	# Line 5
D.	print(nums[x])	# Line 4

Question 46:
You have the following file.

```
index.py:
from sys import argv
print(argv[1] + argv[2])
```

You run the file by executing the following command in the terminal.

```
python index.py 42 3
```

What is the expected output?

A. 423
B. 424242
C. The code is erroneous.
D. 45
E. 126

Question 47:
What is the expected output of the following code?

```
marks = [80, 70, 90, 90, 80, 100]
average = sum(marks) // len(marks)
grade = ''
```

```
if 90 <= average <= 100:
    grade = 'A'
elif 80 <= average < 90:
    grade = 'B'
elif 70 <= average < 80:
    grade = 'C'
elif 65 <= average < 70:
    grade = 'D'
else:
    grade = 'F'

print(grade)
```

A. A
B. F
C. B
D. C
E. D
F. The code is erroneous.

Question 48:
The unnamed except block ...

A. can be placed anywhere.
B. must be the last one.
C. cannot be used if any named block has been used.
D. must be the first one.

Question 49:
What is the expected output of the following code?

```
import random
print(random.seed(3))
```

A. None
B. The code is erroneous.
C. 3
D. None of the above.

Question 50:
Which of the following enclose the input parameters or arguments of a function?

A. Parentheses
B. Quotation marks
C. Curly braces
D. Brackets

Question 51:
Which of the following sentences is true?

```
str1 = 'Peter'
str2 = str1[:]
```

A. str1 and str2 are different (but equal) strings.
B. str1 is longer than str2
C. str1 and str2 are different names of the same string.
D. str2 is longer than str1

Question 52:
You want to be able to read and write data to a file. The file needs to be automatically created, if it doesn't exist. If the file does already exist, you want to override the existing content. Which of the following commands do you have to choose?

A. open('data.txt' , 'w+')
B. open('data.txt' , 'r')
C. open('data.txt' , 'r+')
D. open('data.txt' , 'w')

Question 53:
What is the expected output of the following code?

```
data = ((1, 2),) * 7
print(len(data[3:8]))
```

A. The code is erroneous.
B. 6
C. 4
D. 5

Question 54:
The following line of code ...

```
for line in open('data.txt', 'r'):
```

A. is valid as open returns an iterable object.
B. is invalid as open returns a non-iterable object.
C. may be valid if line is a list.
D. is invalid as open returns nothing.

Question 55:
You are coding a math utility by using Python. You are writing a function to compute roots. The function must meet the following requirements:
If a is non-negative, return a ** (1 / b)
If a is negative and even, return 'Result is an imaginary number'
If a is negaitve and odd, return -(-a) ** (1 / b)
Which of the following functions meets the requirements?

A.

```
def safe_root(a, b):
    if a >= 0:
        answer = -(-a) ** (1 / b)
    elif a % 2 == 0:
        answer = 'Result is an imaginary number'
    else:
        answer = a ** (1 / b)
  return answer
```

B.

```
def safe_root(a, b):
    ifa % 2 == 0:
        answer = -(-a) ** (1 / b)
    elif a >= 0:
        answer = 'Result is an imaginary number'
    else:
        answer = a ** (1 / b)
  return answer
```

C.

```
def safe_root(a, b):
    if a % 2 == 0:
        answer = a ** (1 / b )
    elif a >= 0:
        answer = 'Result is an imaginary number'
    else:
        answer = -(-a) ** (1 / b)
  return answer
```

D.

```
def safe_root(a, b):
    if a >= 0:
        answer = a ** (1 / b)
    elif a % 2 == 0:
        answer = 'Result is an imaginary number'
    else:
        answer = -(-a) ** (1 / b)
  return answer
```

Question 56:
What is the expected output of the following code?

```
data = {'one': 'two', 'two': 'three', 'three': 'one'}
res = data['three']

for _ in range(len(data)):
    res = data[res]

print(res)
```

A. two
B. ('one' , 'two' , 'three')
C. three
D. one

Question 57:
You create a function to calculate the power of a number by using Python. You need to ensure that the function is documented with comments, you create the following code. Line numbers are included for reference only.

```
01 # The calc_power function calculates exponents
02 # x is the base
03 # y is the exponent
04 # The value of x raised to the y power is returned
05 def calc_power(x, y):
06     comment = "# Return the value"
07     return x ** y  # raise x to the y power
```

Which of the following statements are true? Choose two.

A. Line 07 contains an inline comment.
B. Lines 01 through 04 will be ignored for syntax checking.
C. The string in Line 06 will be interpreted as a comment.
D. The pond sign (#) is optional for lines 02 and 03.